

M01 - DATA FOUNDATIONS + ESSENTIAL EXCEL

15-Lesson Canonical Learner Book

The book is complete teaching. External media is support, never a hidden prerequisite.

How every lesson works

START -> OUTCOME -> EXPLAIN -> DEMONSTRATE -> GUIDED REBUILD -> WHY -> CHECK -> RECOVER -> FRESH TRANSFER -> VALIDATE -> HELP -> EXPLAIN BACK -> MASTER -> FINISHED -> NEXT.

DE01-01 - What data is: tables, records, fields and data types

Canonical source role: Support Demonstration

Source: Luke Barousse — Data Engineer Bootcamp (2026)

Assigned segment: 4:16:18 Data Modeling & JOINS + 6:57:49 Data Types

Source state: BASELINE CHECKED 2026-09-08; FINAL SOURCE RECHECK PENDING

[Open assigned source](#)

Fallback: Canonical lesson book + in-course guided lab; no assessment depends on external media.

1. START

Open Raw_Sales. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Understand what a table, row, column, cell and data type are before you touch formulas or SQL.

3. EXPLAIN

Think of a school attendance register. Each student has one row. Name, roll number and attendance are columns. The value where one row and one column meet is a cell.

GLOSSARY

Word Plain-English meaning

Table A rectangular collection of related data.

Row / record One observation or event at the table's current level of detail.

Column / field One attribute stored consistently across rows.

Cell One value at the intersection of a row and column.

Data type The kind of value: text, whole number, decimal, date/time, true/false, etc.

4. DEMONSTRATE

1. Open the workbook sheet Raw_Sales.
2. Look only at the header row first: OrderID, OrderDate, CustomerID, CustomerName, Region, Product, Qty, UnitPrice, PaymentMethod.
3. Pick row 2. Read it left to right as one order record.
4. Classify Qty and UnitPrice as numbers, OrderDate as a date-like value, and Region/Product/PaymentMethod as text categories.

5. WHY

Data tools cannot be used safely until you know what one row and each field mean.

6. CHECK

You should be able to point at any value and say both what column it belongs to and what business thing the entire row represents.

7. RECOVER

- A value looking like a date does not guarantee Excel stored it as a true date.
 - Formatting is appearance; data type is how the value behaves.
 - Do not assume every number should be added. OrderID contains digits but is an identifier, not a quantity.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

G01 - Map table grain and field types

9. FRESH TRANSFER

P01 - Grain and candidate key

In Raw_Sales, choose OrderID, Qty, UnitPrice and PaymentMethod. Write whether each is identifier, numeric measure or text category, and explain why.

10. VALIDATE

You should be able to point at any value and say both what column it belongs to and what business thing the entire row represents.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to Raw_Sales.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer. Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 1 fresh grain/key prompt

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only after Checkpoint 1 fresh grain/key prompt succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-02 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 1 fresh grain/key prompt

DE01-02 - Grain: what one row represents

Canonical source role: Book Led

Source: Canonical lesson + guided lab

External media: Not required

Fallback: Not applicable - the core teaching is in this book and lab.

1. START

Open Raw_Sales + Support_Tickets. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Learn grain - the sentence that says exactly what one row represents.

3. EXPLAIN

A photo can show one person, one family or one whole crowd. A table also has a level of detail. Grain tells you the level.

GLOSSARY

Word Plain-English meaning

Grain The business meaning of one row.

Order header Usually one row per order.

Order line Usually one row per product line inside an order.

Aggregation Combining many lower-level rows into fewer summary rows.

4. DEMONSTRATE

1. Look at Raw_Sales. Ask: if OrderID O1001 appears once, what event does that row describe?
2. Write: One row represents one order record for one customer/product purchase in this practice data.
3. Now imagine an order-line table where one order buys three products. That same OrderID could legitimately appear three times because the grain is one order line, not one order.

5. WHY

Grain controls counting, duplicates, joins and summaries; wrong grain creates believable but wrong answers.

6. CHECK

Before counting, joining or summarizing, you can say "one row represents ..." in plain English.

7. RECOVER

- Saying "grain = customer" just because CustomerID exists.

- Counting rows as orders without proving the table is one row per order.

Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

G01 - Map table grain and field types

9. FRESH TRANSFER

P01 - Grain and candidate key + P02 - Legitimate repeats versus duplicates

For Support_Tickets, write the grain sentence. Then explain why CustomerID may repeat even if TicketID should not.

10. VALIDATE

Before counting, joining or summarizing, you can say "one row represents ..." in plain English.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to Raw_Sales + Support_Tickets.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 1 fresh grain/key prompt

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Checkpoint 1 fresh grain/key prompt succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-03 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 1 fresh grain/key prompt

DE01-03 - Keys, duplicates, NULLs and missing data

Canonical source role: Support Demonstration

Source: Luke Barousse — Data Engineer Bootcamp (2026)

Assigned segment: 4:16:18 Data Modeling & JOINS + 10:16:15 Date/Text/NULL Functions

Source state: BASELINE CHECKED 2026-09-08; FINAL SOURCE RECHECK PENDING

[Open assigned source](#)

Fallback: Canonical lesson book + in-course guided lab; no assessment depends on external media.

1. START

Open Raw_Sales + Support_Tickets. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Tell legitimate repeats from suspicious duplicates and understand missing values.

3. EXPLAIN

A person can buy five times, so their customer ID can repeat. But a single receipt number printed twice for the exact same sale is suspicious.

GLOSSARY

Word Plain-English meaning

Key A column or combination of columns used to identify the row at its intended grain.

Duplicate A repeated row/key that should be unique at that grain.

NULL / blank A missing or unknown value.

Zero A known numeric value equal to nothing - not the same as missing.

4. DEMONSTRATE

1. In Raw_Sales, find the two O1019 rows. Compare every column. They are identical, so this is suspicious at one-order-record grain.
2. Find CustomerID C003 appearing on more than one order. That is not automatically wrong because one customer can place multiple orders.
3. Find the blank CustomerID and blank PaymentMethod. Treat these as missing-information questions, not as zero.

5. WHY

A repeated value is only a duplicate relative to the table grain and key. Missingness also needs a business rule, not automatic deletion.

6. CHECK

You do not delete a repeat until you can explain why it violates the key/grain.

7. RECOVER

- Using "Remove Duplicates" before deciding the key.

- Replacing blanks with 0 just to make them disappear.

Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

G02 - Key and duplicate inspection + G03 - Missingness and validity scan

9. FRESH TRANSFER

P02 - Legitimate repeats versus duplicates

In Raw_Sales, write: candidate key, one legitimate repeat, one suspicious duplicate, and two missing-value issues.

10. VALIDATE

You do not delete a repeat until you can explain why it violates the key/grain.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to Raw_Sales + Support_Tickets.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 1 duplicate/missing reasoning

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Checkpoint 1 duplicate/missing reasoning succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-04 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 1 duplicate/missing reasoning

DE01-04 - Data quality dimensions and validation thinking

Canonical source role: Support Demonstration

Source: Statistics Canada — Data Quality in Six Dimensions

Assigned segment: Full 8:42 beginner video

Source state: CURRENT WEB CHECK 2026-09-09

[Open assigned source](#)

Fallback: Canonical lesson book + in-course guided lab; no assessment depends on external media.

1. START

Open Raw_Sales. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Turn "data quality" from a vague phrase into specific checks.

3. EXPLAIN

Before accepting a food delivery you might check: did everything arrive, is anything broken, is it the right item, did you receive it on time? Data quality works the same way.

GLOSSARY

Word Plain-English meaning

Completeness Are required values present?

Validity Do values obey allowed rules/ranges?

Uniqueness Are values that should be unique actually unique?

Consistency Do the same concepts use consistent labels/formats?

Timeliness Is the data recent enough for the decision?

4. DEMONSTRATE

1. Completeness: check CustomerID/PaymentMethod blanks.
2. Validity: Qty should normally be positive; Region should belong to the approved set.
3. Uniqueness: OrderID should be unique if grain is one order.
4. Consistency: East and east should not silently become different business regions.
5. Timeliness: an old dataset may be correct but too stale for today's operational decision.

5. WHY

Quality checks turn vague trust into explicit tests that can fail visibly before bad data reaches reporting.

6. CHECK

Every quality claim becomes a concrete test: column + rule + what failure looks like.

7. RECOVER

- Writing "check data quality" without a test.

- Confusing uniqueness with "no repeated values anywhere".

Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

G03 - Missingness and validity scan

9. FRESH TRANSFER

P03 - Design five data-quality checks

Using Raw_Sales, write one specific check for each of the five dimensions.

10. VALIDATE

Every quality claim becomes a concrete test: column + rule + what failure looks like.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to Raw_Sales.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 1 concrete quality-test design

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Checkpoint 1 concrete quality-test design succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-05 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 1 concrete quality-test design

DE01-05 - Meet Excel: workbook, worksheet, cells and formulas

Canonical source role: Primary Video

Source: Chandoo — Excel Tutorial for Beginners

Assigned segment: 1:15–4:09

Source state: CURRENT SEARCH RESOLUTION 2026-09-09

[Open assigned source](#)

Fallback: Canonical lesson book + in-course guided lab; no assessment depends on external media.

1. START

Open START_HERE + workbook working copy. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Understand the Excel screen: workbook, worksheet, cell, address, formula bar and formula.

3. EXPLAIN

A workbook is like a notebook file. Worksheets are the tabs/pages inside it. A cell is one box on a page.

GLOSSARY

Word Plain-English meaning

Workbook The .xlsx file.

Worksheet A tab inside the workbook.

Cell address Column letter + row number, such as D2.

Formula bar The area that shows the actual value/formula stored in the active cell.

Formula An instruction beginning with = that Excel calculates.

4. DEMONSTRATE

1. Open M01_EXCEL_PRACTICE_WORKBOOK_TEACHING_RC2.xlsx.
2. Click the Formula_Practice tab.
3. Click cell D2. Notice the Name Box/selected cell address says D2.
4. Type =B2*C2 and press Enter.
5. Click D2 again. The cell shows a result, while the formula bar shows =B2*C2.

5. WHY

Excel instructions depend on precise workbook, worksheet and cell language; navigation mistakes later become formula mistakes.

6. CHECK

You understand the difference between what the cell displays and the formula stored behind it.

7. RECOVER

- Typing 1700 manually instead of =B2*C2.

- Typing B2*C2 without the leading =.

Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

G04 - Workbook navigation proof

9. FRESH TRANSFER

Checkpoint 2 - navigation/formula setup proof

Click three different cells and say their addresses aloud. Then write one simple addition formula in an unused cell and undo it.

10. VALIDATE

You understand the difference between what the cell displays and the formula stored behind it.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to START_HERE + workbook working copy.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 2 closed-book workbook navigation

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Checkpoint 2 closed-book workbook navigation succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-06 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 2 closed-book workbook navigation

DE01-06 - Enter, save and structure tabular data safely

Canonical source role: Primary Video

Source: Chandoo — Excel Tutorial for Beginners

Assigned segment: 4:09–8:21

Source state: CURRENT SEARCH RESOLUTION 2026-09-09

[Open assigned source](#)

Fallback: Canonical lesson book + in-course guided lab; no assessment depends on external media.

1. START

Open Raw_Sales working copy. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Enter and save tabular data without destroying the original source.

3. EXPLAIN

A raw file is like an original photograph. Edit a copy so you can always return to the untouched original.

GLOSSARY

Word Plain-English meaning

Raw data Original received data before your transformations.

Working copy A separate file you are allowed to edit.

Header row The row containing column names.

Tabular structure One header row, one observation per row, one attribute per column.

4. DEMONSTRATE

1. Open the workbook. Use File > Save As and create a working copy such as M01_working.xlsx.
2. Keep one header row. Avoid merged cells inside data tables.
3. Do not insert decorative blank rows in the middle of the dataset.
4. Save frequently with Ctrl+S.

5. WHY

Preserving the source and using a working copy makes mistakes reversible and keeps evidence reproducible.

6. CHECK

The original workbook remains unchanged, and your working copy contains your experiments.

7. RECOVER

- Editing the only copy of the raw file.

- Using two header rows or merged cells as decoration inside source data.

Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

G04 - Workbook navigation proof + safe working-copy discipline

9. FRESH TRANSFER

Checkpoint 2 - safe working-copy and structure proof

Create a working copy, rename it clearly, and write the full filename in your evidence notes.

10. VALIDATE

The original workbook remains unchanged, and your working copy contains your experiments.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to Raw_Sales working copy.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 2 preserve source + working-copy proof

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Checkpoint 2 preserve source + working-copy proof succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-07 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 2 preserve source + working-copy proof

DE01-07 - Sort and filter without damaging data

Canonical source role: Primary Video

Source: Chandoo — Excel Tutorial for Beginners

Assigned segment: 11:37–15:46

Source state: CURRENT SEARCH RESOLUTION 2026-09-09

[Open assigned source](#)

Fallback: Canonical lesson book + in-course guided lab; no assessment depends on external media.

1. START

Open Raw_Sales working copy. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Use Sort and Filter as viewing/ordering tools without deleting or misaligning data.

3. EXPLAIN

A filter is like temporarily hiding books that do not match your search; the books are still on the shelf.

GLOSSARY

Word Plain-English meaning

Filter Temporarily shows rows matching conditions; hidden rows remain in the table.

Sort Changes row order based on one or more columns.

Whole-row integrity Values from the same record must move together.

4. DEMONSTRATE

1. In a working copy of Raw_Sales, click inside the data.
2. Choose Data > Filter.
3. Filter Region to East/east and note the visible rows.
4. Clear the filter and confirm the full set returns.
5. For sorting, click inside the table and sort the entire data range by UnitPrice - do not select only UnitPrice cells in isolation.

5. WHY

Filtering should change only what you see; sorting should change order without breaking row integrity.

6. CHECK

Filtering never deletes the hidden rows; sorting keeps OrderID, customer, product, Qty and price aligned on the same record.

7. RECOVER

- Deleting hidden rows because they are not visible.

- Sorting one column by itself and scrambling row relationships.

Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

G05 - Safe sort/filter

9. FRESH TRANSFER

P04 - Filter and prove no deletion

Filter one PaymentMethod, note the visible count, clear it, and confirm the original row count returns.

10. VALIDATE

Filtering never deletes the hidden rows; sorting keeps OrderID, customer, product, Qty and price aligned on the same record.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to Raw_Sales working copy.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 2 filter/restore + safe sort

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Checkpoint 2 filter/restore + safe sort succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-08 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 2 filter/restore + safe sort

DE01-08 - Basic formulas and essential functions

Canonical source role: Primary Video

Source: Chandoo — Excel Tutorial for Beginners

Assigned segment: 15:46–22:39

Source state: CURRENT SEARCH RESOLUTION 2026-09-09

[Open assigned source](#)

Fallback: Canonical lesson book + in-course guided lab; no assessment depends on external media.

1. START

Open Formula_Practice. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Create basic formulas and use SUM, AVERAGE and COUNT/COUNTA.

3. EXPLAIN

A formula is a recipe. If the ingredients change, Excel should recalculate the result automatically.

GLOSSARY

Word Plain-English meaning

Cell reference A pointer to a cell such as B2.

SUM Adds numeric values.

AVERAGE Calculates arithmetic mean.

COUNT Counts numeric cells.

COUNTA Counts nonblank cells.

4. DEMONSTRATE

1. Open Formula_Practice. In D2 type =B2*C2.
2. Fill/copy D2 down through the product rows.
3. Manually check Keyboard: $2 \times 850 = 1700$.
4. Use =SUM(D2:D7) for total Sales.
5. Use =AVERAGE(D2:D7) for average Sales.
6. Use COUNT on a numeric column or COUNTA on a populated text/ID column depending on what you intend to count.

5. WHY

Formulas make calculations repeatable and auditable; manual arithmetic is used as an independent control, not as the production method.

6. CHECK

Changing a Qty or UnitPrice automatically changes the calculated Sales; formulas appear in the formula bar.

7. RECOVER

- Typing answers instead of formulas.

- Using COUNT on text IDs and thinking zero means there are no records.

Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

G06 - Basic row formulas

9. FRESH TRANSFER

P05 - Revenue and Margin formulas

Complete Formula_Practice and manually validate two rows before trusting the total.

10. VALIDATE

Changing a Qty or UnitPrice automatically changes the calculated Sales; formulas appear in the formula bar.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to Formula_Practice.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 2 fresh Sales formula + manual check

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Checkpoint 2 fresh Sales formula + manual check succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-09 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 2 fresh Sales formula + manual check

DE01-09 - Relative and absolute references

Canonical source role: Primary Video

Source: Chandoo — Useful Excel concepts

Assigned segment: 12:35–20:52

Source state: CURRENT SEARCH RESOLUTION 2026-09-09

[Open assigned source](#)

Fallback: Canonical lesson book + in-course guided lab; no assessment depends on external media.

1. START

Open Reference_Practice. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Understand why A2 moves when copied but \$H\$2 stays fixed.

3. EXPLAIN

A relative reference is like "the house next door" - it changes when you move. An absolute reference is like a full GPS coordinate - it stays fixed.

GLOSSARY

Word Plain-English meaning

Relative reference A2; changes relative to the formula location when copied.

Absolute reference \$H\$2; row and column are locked.

Mixed reference \$H2 or H\$2; only one part is locked.

4. DEMONSTRATE

1. Open Reference_Practice. Calculate Base Sales in D2 using Qty x UnitPrice.
2. Tax Rate is stored once in H2. In E2 use a formula such as $=D2*(1+\$H\$2)$.
3. Copy E2 down. Click E3, E4, E5. D2 changes to D3/D4/D5, but \$H\$2 stays exactly \$H\$2.
4. Temporarily change H2. Every taxed result should update because all formulas point to the same rate cell. Undo if needed.

5. WHY

Relative references should move with rows; constants such as a tax/rate cell must stay fixed when copied.

6. CHECK

One setting cell controls every row without repeating the percentage in each formula.

7. RECOVER

- Using H2 without dollar signs and watching it drift to H3/H4.

- Typing 18% directly in every row formula.

Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

G07 - Relative vs absolute reference

9. FRESH TRANSFER

P06 - One absolute rate cell

Complete Independent_Rate using the single rate in H2 and explain what each dollar sign protects.

10. VALIDATE

One setting cell controls every row without repeating the percentage in each formula.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to Reference_Practice.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 3 fresh absolute-reference task

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Checkpoint 3 fresh absolute-reference task succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-10 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 3 fresh absolute-reference task

DE01-10 - Excel Tables and structured data habits

Canonical source role: Primary Video

Source: Chandoo — Useful Excel concepts

Assigned segment: 2:22–8:45

Source state: CURRENT SEARCH RESOLUTION 2026-09-09

[Open assigned source](#)

Fallback: Canonical lesson book + in-course guided lab; no assessment depends on external media.

1. START

Open clean working range. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Know the difference between a normal range and an actual Excel Table.

3. EXPLAIN

A normal range is a group of cells. An Excel Table is a managed object that knows its headers, rows and filters.

GLOSSARY

Word Plain-English meaning

Excel Table A structured Excel object created with Insert > Table or Ctrl+T.

Header Named field at the top of a Table column.

Structured behavior Filters, automatic expansion and formula propagation tied to the Table object.

4. DEMONSTRATE

1. Work on a clean copy/range. Click inside it.
2. Choose Insert > Table (or Ctrl+T).
3. Confirm the full range and tick My table has headers.
4. Notice filter arrows appear. Add one new row directly below the Table and see the Table expand.

5. WHY

Excel Tables create explicit rectangular structure, stable headers and predictable expansion/filter behavior.

6. CHECK

The object expands with new rows and keeps explicit headers/filters.

7. RECOVER

- Assuming coloured formatting means the data is an Excel Table.

- Creating a Table from an incomplete range that omits columns.

Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

G08 - Excel Table behaviour

9. FRESH TRANSFER

P07 - Prove Excel Table behaviour

Convert a small clean range to a Table, add a row, then describe one behavior that proves it is a Table.

10. VALIDATE

The object expands with new rows and keeps explicit headers/filters.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to clean working range.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 3 fresh Table behavior proof

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Checkpoint 3 fresh Table behavior proof succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-11 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 3 fresh Table behavior proof

DE01-11 - Practical cleaning and lookup awareness

Canonical source role: Support Demonstration

Source: Chandoo — XLOOKUP for Beginners

Assigned segment: Whole short video

Source state: BASELINE CHECKED 2026-09-08; FINAL SOURCE RECHECK PENDING

[Open assigned source](#)

Fallback: Canonical lesson book + in-course guided lab; no assessment depends on external media.

1. START

Open Raw_Sales / PowerQuery_Source + Customer_Lookup. Keep original/source material unchanged unless the task explicitly

says to edit a working copy.

2. OUTCOME

Understand basic cleaning and how XLOOKUP connects related tables using an ID.

3. EXPLAIN

A lookup is like using a roll number to find a student's house from a separate register. The ID is the matching key.

GLOSSARY

Word Plain-English meaning

Cleaning Making values consistent/usable without changing their business meaning.

Lookup value The ID you are trying to find.

Lookup array Where Excel searches for that ID.

Return array The column from which Excel returns the answer.

Not found The explicit result when no match exists.

4. DEMONSTRATE

1. Notice Region values East, east and West with trailing space in Raw_Sales. These are consistency problems.

2. Open Customer_Lookup. CustomerID is the linking field; Tier is the value you want to bring back.

3. In Independent_Lookup, use XLOOKUP with CustomerID as the lookup value, Customer_Lookup!CustomerID as search range, and Tier as return range.

4. Handle C999 explicitly with a not-found message rather than silently assuming a tier.

5. WHY

Cleaning should standardize representation without inventing business meaning; lookups connect tables by a deliberate key.

6. CHECK

Known IDs return the same Tier shown in Customer_Lookup; C999 is clearly flagged missing.

7. RECOVER

- Matching by CustomerName when CustomerID is the intended key.

- Treating unmatched IDs as blank and forgetting to investigate them.

Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

G09 - Cleaning awareness + lookup follow-along in learner book

9. FRESH TRANSFER

P08 - Lookup Tier by CustomerID

Complete Independent_Lookup, spot-check two known customers, and explain the four pieces of XLOOKUP in plain English.

10. VALIDATE

Known IDs return the same Tier shown in Customer_Lookup; C999 is clearly flagged missing.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to Raw_Sales / PowerQuery_Source + Customer_Lookup.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 3 fresh lookup with missing-ID handling

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Checkpoint 3 fresh lookup with missing-ID handling succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-12 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 3 fresh lookup with missing-ID handling

DE01-12 - PivotTable and Power Query awareness

Canonical source role: Support Demonstration

Source: Chandoo — PivotTable / Power Query awareness

Assigned segment: 31:35–38:35

Source state: CURRENT SEARCH RESOLUTION 2026-09-09

[Open assigned source](#)

Fallback: Canonical lesson book + in-course guided lab; no assessment depends on external media.

1. START

Open Pivot_Source + PowerQuery_Source. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Understand what PivotTables and Power Query are for without trying to master them yet.

3. EXPLAIN

PivotTable is a fast summary machine. Power Query is a recorded cleaning/import recipe you can replay.

GLOSSARY

Word Plain-English meaning

PivotTable Interactive summary built from source data.

Rows area Categories listed vertically.

Values area Measures summarized, e.g., Sum of Sales.

Power Query Excel's repeatable import/transform tool.

Refresh Run the saved import/transform logic again on updated data.

4. DEMONSTRATE

1. Pivot: click Pivot_Source > Insert > PivotTable > New Worksheet. Put Region in Rows and Sales in Values.
2. Check that Values says Sum of Sales, not Count of Sales.
3. Power Query awareness: inspect PowerQuery_Source. Find leading/trailing spaces, inconsistent case, mixed date formats and Amount stored as text with comma.
4. The point is not to hand-fix every cell. A Power Query workflow can record trim/clean/type/standardization steps and refresh them later.

5. WHY

PivotTables summarize quickly; Power Query records repeatable import/cleaning logic. At this stage you need a safe mental model, not advanced BI mastery.

6. CHECK

You can answer which Region has the highest sales and describe Power Query as repeatable transformation rather than "another worksheet".

7. RECOVER

- Building a fancy dashboard before understanding the summary.
 - Using Count of Sales by accident.
 - Manually cleaning source values one-by-one and calling that a repeatable pipeline.
- Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

Pivot follow-along in learner book + P09 after guided attempt

9. FRESH TRANSFER

P09 - Pivot summary and reconciliation

Create the simple PivotTable, reconcile its grand total to source SUM, then write 2-3 sentences explaining Power Query.

10. VALIDATE

You can answer which Region has the highest sales and describe Power Query as repeatable transformation rather than "another worksheet".

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to Pivot_Source + PowerQuery_Source.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 4 fresh Pivot summary + source reconciliation

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Checkpoint 4 fresh Pivot summary + source reconciliation succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-13 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 4 fresh Pivot summary + source reconciliation

DE01-13 - Chart choice and basic visual communication

Canonical source role: Support Demonstration

Source: Chandoo — What is a Chart and How to Make Charts in Excel (video)

Assigned segment: Embedded beginner video: chart purpose, create/format/change chart type; use the page's "select correct chart" follow-up only as needed

Source state: CURRENT WEB CHECK 2026-09-09

[Open assigned source](#)

Fallback: Canonical lesson book + in-course guided lab; no assessment depends on external media.

1. START

Open Pivot_Source / summary range. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Choose a simple chart that matches the question and avoid decorative charts that hide the message.

3. EXPLAIN

A chart is a sentence in picture form. Choose the picture that makes the comparison easiest to read.

GLOSSARY

Word Plain-English meaning

Bar/column chart Good for comparing categories.

Line chart Good for trends over ordered time.

Pie chart Limited use; difficult when many categories/slices exist.

Axis Scale along horizontal/vertical direction.

Title Should state what is being compared, not just "Chart 1".

4. DEMONSTRATE

1. Use the regional summary from your PivotTable.
2. For comparing East/West/North/South, choose a bar or column chart.
3. Give it a clear title such as "Sales by Region".
4. Check that the largest bar visually matches the largest numeric total.

5. WHY

A chart should answer the business question clearly; decorative visuals can hide the comparison instead of explaining it.

6. CHECK

A reader can answer the business question quickly without decoding unnecessary colours or 3-D effects.

7. RECOVER

- Using a line chart for unordered categories.
- Adding 3-D effects that distort comparisons.
- Making the title vague.

Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

Simple chart follow-along + Checkpoint 4 visual task

9. FRESH TRANSFER

Checkpoint 4 - create a simple comparison chart

Create one simple regional comparison chart and write one sentence explaining why that chart type fits the question.

10. VALIDATE

A reader can answer the business question quickly without decoding unnecessary colours or 3-D effects.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to Pivot_Source / summary range.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 4 fresh chart-choice task

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Checkpoint 4 fresh chart-choice task succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-14 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 4 fresh chart-choice task

DE01-14 - Reconciliation, sanity checks and source preservation

Canonical source role: Support Demonstration

Source: Posit Open Source — pointblank data validation workshop

Assigned segment: Use the validation/reporting + table row-count matching portions; full workshop is 1h53m and may be watched as a long class

Source state: CURRENT WEB CHECK 2026-09-09

[Open assigned source](#)

Fallback: Canonical lesson book + in-course guided lab; no assessment depends on external media.

1. START

Open completed working copy + source data. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Learn to distrust even plausible-looking results until you reconcile them to an independent check.

3. EXPLAIN

If a shop cashier says the total is 3810, you may still count the item totals separately. Reconciliation is that second route to the answer.

GLOSSARY

Word Plain-English meaning

Reconciliation Comparing the result to an independent total/count/control.

Source preservation Keeping raw evidence unchanged.

Sanity check A quick reasonableness check for impossible or surprising values.

Control total A known count/amount used to verify processing.

4. DEMONSTRATE

1. Before transformations, record source row count and any important total.
2. After filtering/cleaning/calculating, recompute the total another way.
3. If Pivot grand total differs from SUM of source Sales, stop and find the earliest step where rows/values diverged.
4. Do not "fix the final number" manually; repair the logic.

5. WHY

Plausible output is not proof. Reconciliation and source preservation make a result independently checkable and recoverable.

6. CHECK

You can show both the result and one independent check that supports it.

7. RECOVER

- Trusting a pretty PivotTable because it looks professional.

- Changing raw data to force agreement.

Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

Reconciliation follow-along + Checkpoint 4

9. FRESH TRANSFER

P09 + Checkpoint 4 - reconcile to independent total

For Pivot_Source, compare Pivot grand total with =SUM(E2:E19). Record both and explain whether they reconcile.

10. VALIDATE

You can show both the result and one independent check that supports it.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to completed working copy + source data.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Checkpoint 4 independent reconciliation + limitation

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Checkpoint 4 independent reconciliation + limitation succeeds without the worked lesson/review open.

15. NEXT ROUTE

Proceed to DE01-15 only if mastery evidence is stable.

ASSESSMENT ROUTE

Checkpoint 4 independent reconciliation + limitation

DE01-15 - Mini workflow: inspect -> calculate -> validate -> explain ->

Canonical source role: Book Led
Source: Canonical lesson + guided lab
External media: Not required
Fallback: Not applicable - the core teaching is in this book and lab.

1. START

Open one completed M01 working file. Keep original/source material unchanged unless the task explicitly says to edit a working copy.

2. OUTCOME

Combine the module into a repeatable beginner workflow: inspect -> calculate -> validate -> explain -> save evidence.

3. EXPLAIN

A good analyst is not someone who can press many Excel buttons. It is someone who can explain what data they received, what they changed, what result they produced, and why they trust it.

GLOSSARY

Word Plain-English meaning

Inspect Understand columns, grain, keys and quality before calculating.

Calculate Apply formulas/summaries that answer the question.

Validate Use an independent check.

Explain State result and reasoning in plain language.

Evidence Saved workbook, note, screenshot, formula or output proving what you actually did.

4. DEMONSTRATE

1. Read the business question first.
2. State the table grain and candidate key.
3. Preserve raw source and work on a copy.
4. Perform only the transformations/calculations needed.
5. Validate with row counts, control totals or spot checks.
6. Explain the answer, one limitation and one possible failure mode.
7. Save the workbook and note where the evidence lives.

5. WHY

A professional workflow is a chain: inspect, calculate, validate, explain and save evidence. Skipping any link makes handoff fragile.

6. CHECK

Someone else can reproduce and audit your work instead of trusting an unexplained final number.

7. RECOVER

- Starting with a chart/formula before understanding the data.
- Saving only screenshots and losing the actual workbook/formulas.
- Reporting "done" with no validation.

Recovery rule: stop at the earliest step where your result differs; repair that step, rerun the checkpoint, and only then

continue.

BUILD -> TRANSFER -> PROVE

8. GUIDED REBUILD

Module workflow rehearsal before practical

9. FRESH TRANSFER

P10 - Mini analyst handoff + Module Practical

Write a 5-8 sentence handoff for one M01 practice task: source, grain, what you changed, result, validation, limitation.

10. VALIDATE

Someone else can reproduce and audit your work instead of trusting an unexplained final number.

Independent validation: save at least one count, spot-check, manual calculation, source comparison, or reconciliation appropriate to the task.

11. HELP LADDER

Hint 1 - Concept: revisit the lesson mental model.

Hint 2 - Location: return to one completed M01 working file.

Hint 3 - Partial skeleton: reproduce only the first demonstrated step, then finish without reopening the worked answer.

Review opens only after a genuine attempt; mastery still needs a fresh retry if the attempt was weak.

12. EXPLAIN BACK

Explain the main idea in 30-60 seconds without reading the paragraph above. If you cannot explain it yet, revisit the worked example before moving on.

13. MASTER

Module Practical; if weak, different-domain Fresh Retest

14. FINISHED WHEN

Save the actual workbook/notes, one independent validation, confidence 0-5, and the biggest uncertainty. Mark mastered only

after Module Practical; if weak, different-domain Fresh Retest succeeds without the worked lesson/review open.

15. NEXT ROUTE

Attempt the M01 Module Practical. If weak, perform targeted repair and use the different-domain Fresh Retest; then Gate 1.

ASSESSMENT ROUTE

Module Practical; if weak, different-domain Fresh Retest